FOR SPACES FINITELY PRESENTED OVER A RING, TESTING AGAINST PERFECT COMPLEXES ONLY IS ENOUGH TO DETECT RELATIVE PSEUDO-COHERENCE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 12229-12237 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 12239-12248, the complete author proof (8 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Cites lemma-characterize-pseudo-coherent whose card is held pending the Chief's side ruling; the citation is in-source text and does not depend on that card's registration.

Editorial note. Counted B side per ruling (same label family as the L16205 conflict but its own distinct label).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Derived Categories of Spaces, proposition detecting-bounded-above and lemma perfect-enough; in-source lemma-characterize-pseudo-coherent (the pending-ruling companion card).

Source reference. Original lemma 0CSL: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12165–12172. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CWH: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12248–12248. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GFH: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12248–12248. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let A be a ring. Let X be an algebraic space separated and of finite presentation over A. Let $K \in D_{QCoh}(\mathcal{O}_X)$. If $R\Gamma(X, E \otimes^{\mathbf{L}} K)$ is pseudo-coherent in $D(A)$ for every perfect $E \in D(\mathcal{O}_X)$, then K is pseudo-coherent relative to A.*

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. In view of Lemma 0CSL, it suffices to show $R\Gamma(X, E \otimes^{\mathbf{L}} K)$ is pseudo-coherent in $D(A)$ for every pseudo-coherent $E \in D(\mathcal{O}_X)$. By Derived Categories of Spaces, Proposition 0GFH it follows that $K \in D^-_{QCoh}(\mathcal{O}_X)$. Now the result follows by Derived Categories of Spaces, Lemma 0CWH. □